



भारतीय सर्वेक्षण विभाग
Survey of India
विज्ञान और प्रौद्योगिकी विभाग
Department of Science & Technology

STANDARD BENCH MARKS

**National Institute for Geo-Informatics Science & Technology,
Hyderabad**

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1 Introduction

Bench marks are fundamental reference points in surveying that provide precise elevation data. The Survey of India maintains various types of standard bench marks across the country to establish a unified vertical datum. This document classifies and describes the different types of standard bench marks used in India.

2 Standard Bench Mark Type 'P'

Important Note

Type 'P' bench marks are generally erected/constructed in big towns or at junctions of High-Precision (HP) levelling lines. Only high-quality stone should be used for the monolith, with site selection assistance from the Director of Geological Survey.

Construction Details

- **Monolith:** 0.60m square at base, 0.90m high
- **Top Section:** Upper 0.10m dressed to form a frustum of a pyramid terminating in a square of 0.08m side
- **Foundation:** Rests on a concrete bed 2m square, 0.75m deep
- **Facing:** Stone or brick masonry 1.50m square and 0.30m deep
- **Height:** Upper surface of monolith about 0.60m above ground level
- **Inscription:** "G.T.S. Standard Bench-Mark (year of construction)" engraved on one face



Figure 1: Photograph of Type 'P' Standard Bench Mark



Figure 2: Type 'P' Standard Bench Mark

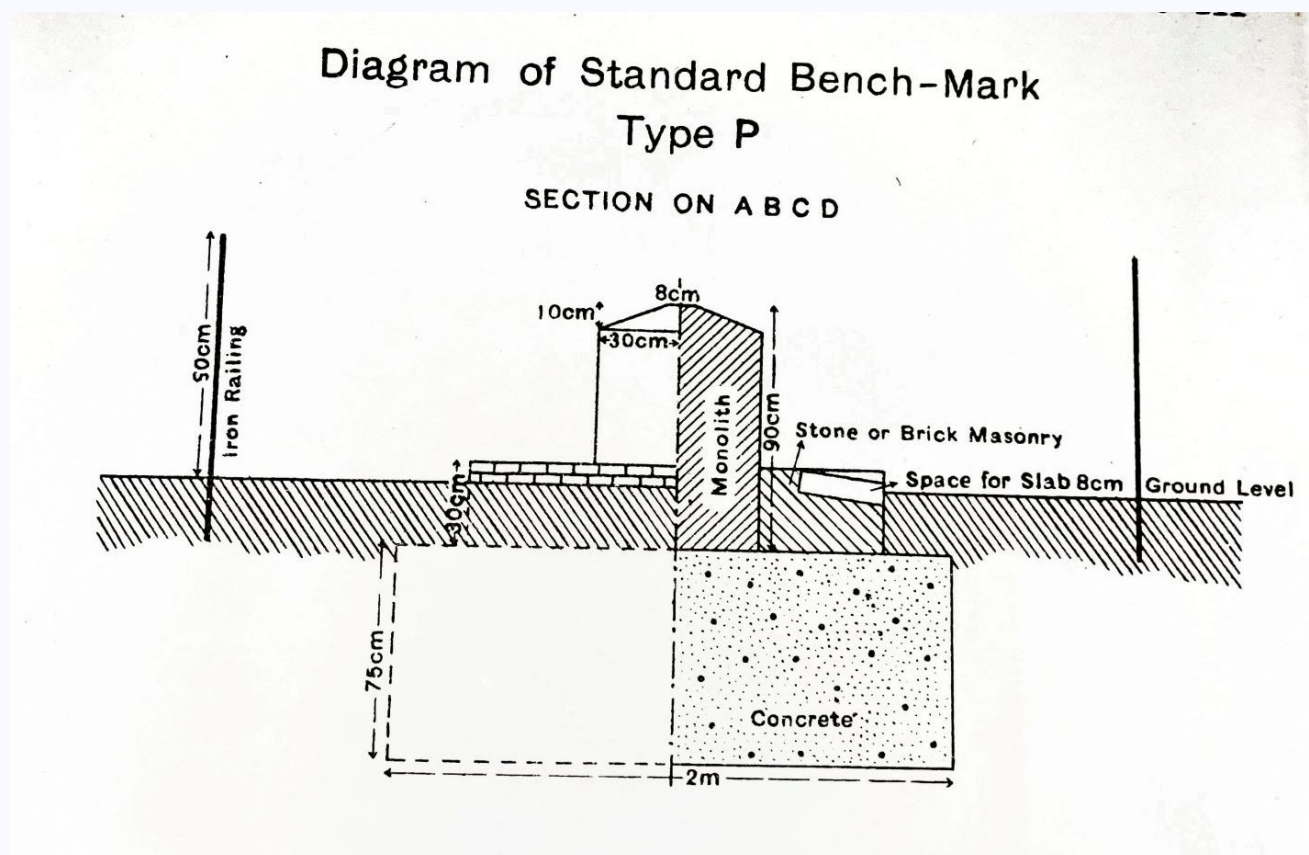


Figure 3: Diagram of Type 'P' Standard Bench Mark

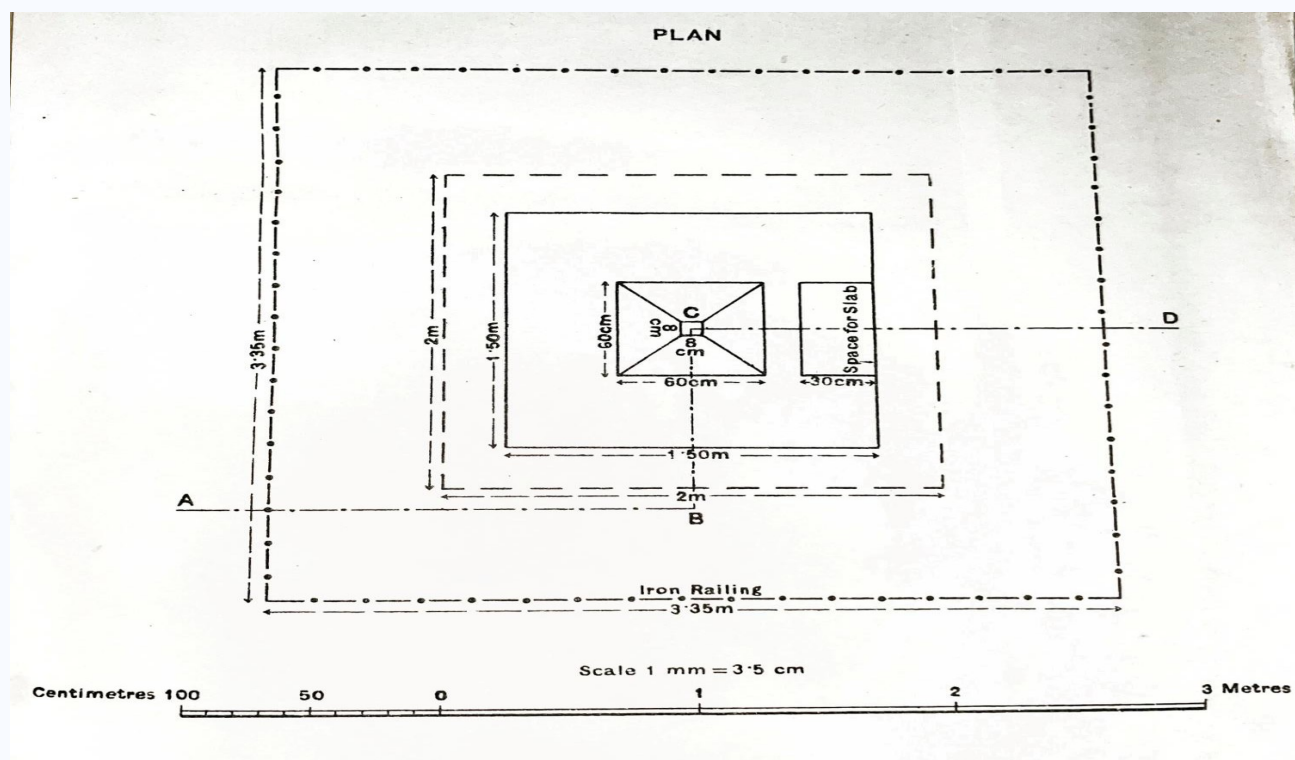


Figure 4: Sketch of Type 'P' Standard Bench Mark

3 Standard Bench-Mark Type 'M'

Important Note

Type 'M' bench marks are constructed in smaller towns and at junctions of HP levelling lines to reduce expense while maintaining precision.

Construction Details

- **Monolith:** Chunār sandstone, 0.30m square and 0.85m high
- **Exposed portion:** 0.25m well-dressed above ground level
- **Embedded portion:** 0.60m undressed, embedded in concrete bed 1.50m square and 0.90m deep
- **Top:** 4cm dressed to form a frustum of a pyramid terminating in a square of 8cm side
- **Enclosure:** Masonry or brick wall 1.50m square outside, 0.30m thick and 0.75m high
- **Supplementary marks:** Two hard stone prisms (0.15m square) embedded flush with cement bed, and one iron bolt fixed horizontally in the enclosure wall



Figure 5: Type 'M' Standard Bench Mark



Figure 6: Photograph of Type 'M' Standard Bench Mark

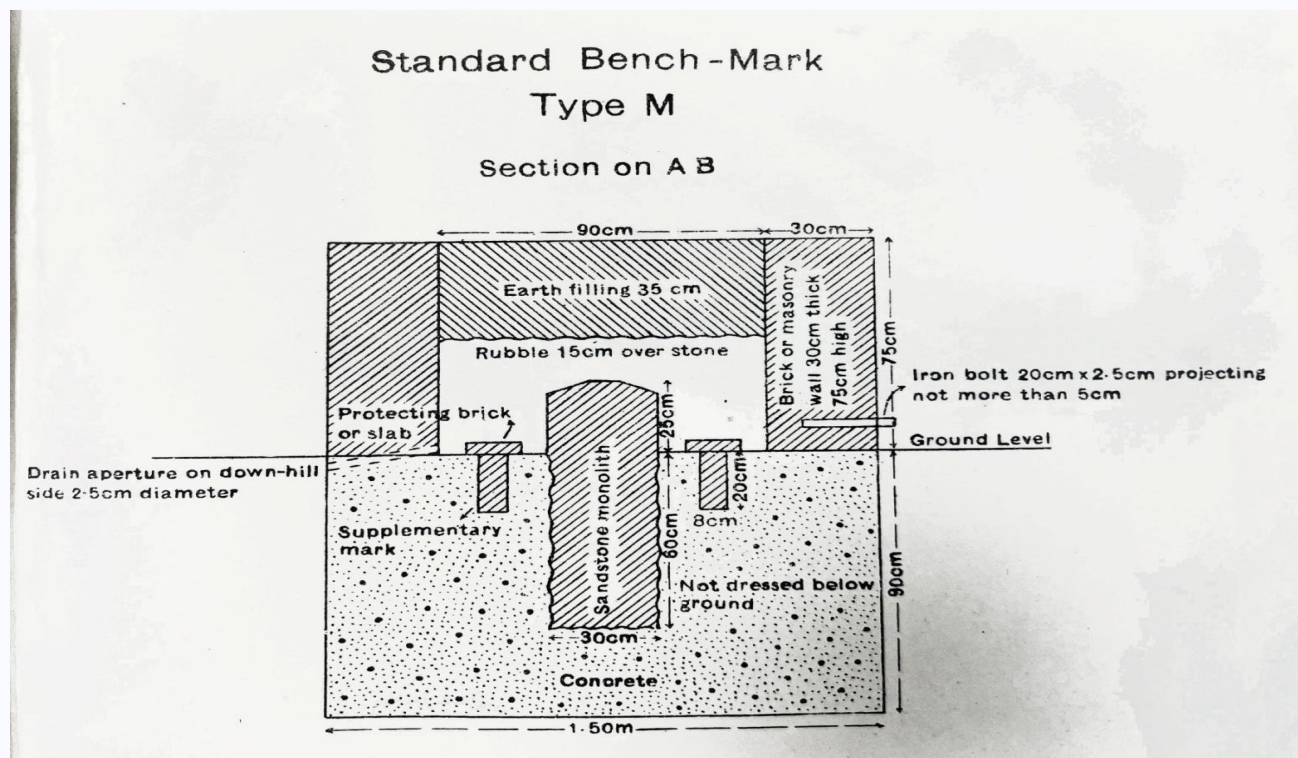


Figure 7: Diagram of Type 'M' Standard Bench Mark

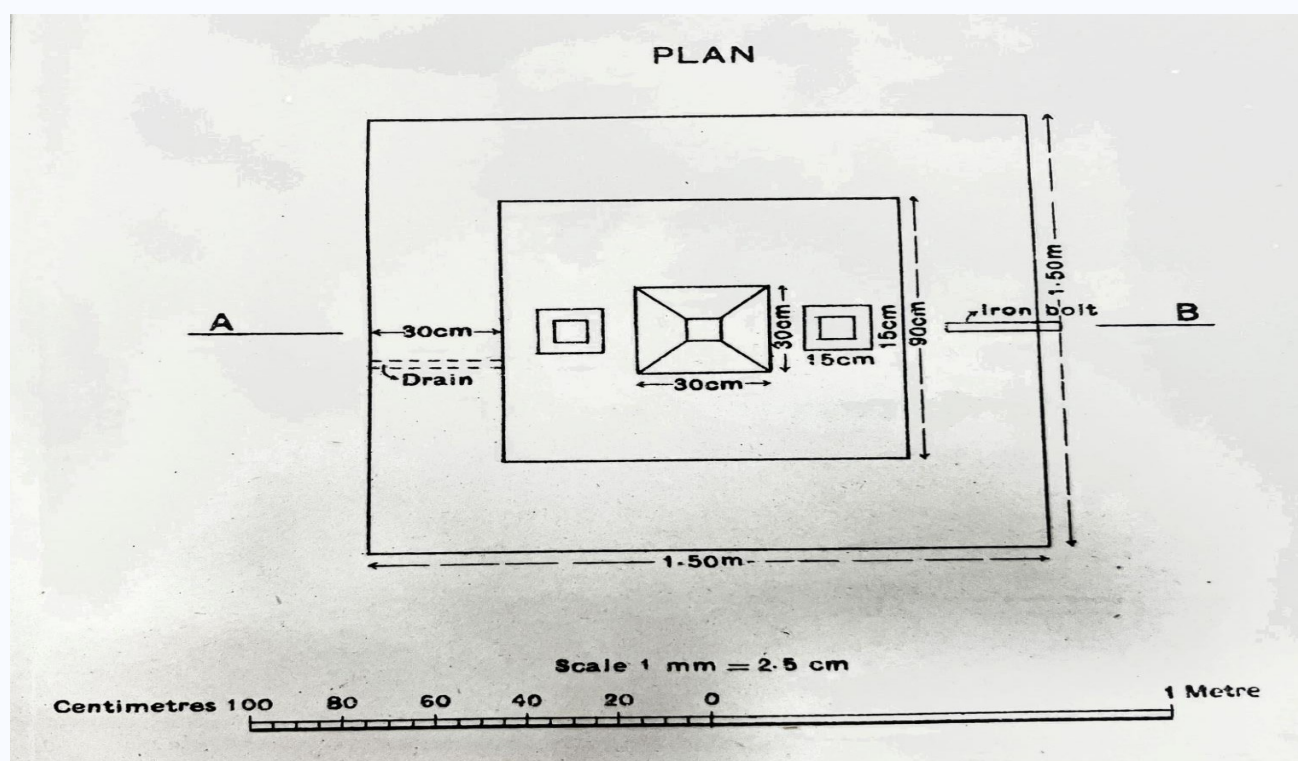


Figure 8: Sketch of Type 'M' Standard Bench Mark

4 Standard Bench Mark Type 'A'

Construction Details

- **Pipe:** Galvanized iron pipe, 0.10m diameter, 0.90m long
- **Foundation:** Concrete slab 0.30m square, 5cm thick
- **Installation:** Top of pipe about 0.55m below ground surface
- **Cap:** Solid zinc cap screwed and riveted to pipe end with inscription
- **Protection:** Brick vault with concrete foundations
- **Aperture:** 0.15m square aperture in vault crown with hollow masonry pillar above
- **Mark stone:** Bears inscription "G.T. Survey O Bench Mark"



Figure 9: Type 'A' Standard Bench Mark

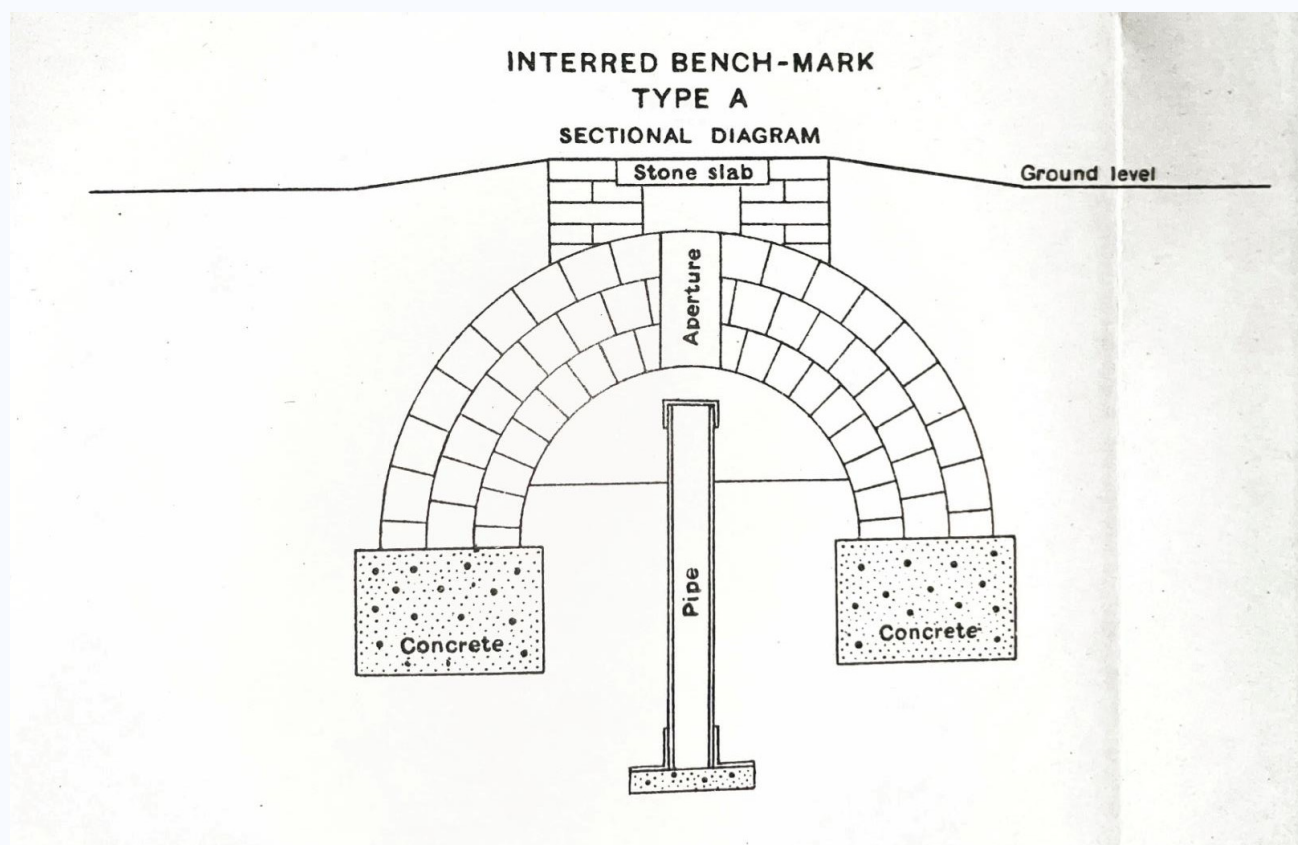


Figure 10: Diagram of Type 'A' Standard Bench Mark

5 Standard Bench Mark Type 'B'

Important Note

Constructed in Gangetic plain areas where natural rock is unavailable, using dressed monolith of Chunar stone flushed with cement concrete bed.

Construction Details

- **Stone:** Dressed stone $0.45\text{m} \times 0.30\text{m} \times 0.30\text{m}$
- **Foundation:** Concrete bed 0.90m square, 0.60m thick
- **Installation:** Upper surface flush with brickwork, about 0.60m below ground level
- **Inscription:** "G.T.S. □ B.M. A.D. (Year of Construction)"
- **Central mark:** 0.15m square, 6mm deep

5.1 Reference Pillar

Construction Details

- **Location:** 2.0m from bench mark
- **Dimensions:** 0.90m high frustum of pyramid, 0.60m square base tapering to 0.15m square top
- **Inscription:** "G T S B M 2.0 m" on face opposite bench mark

Important Note

When dressed stone unavailable, use cemented concrete slab with brass tablet insert (9cm diameter disk with 2cm thick, 9cm long rod below)



Figure 11: Type 'B' Standard Bench Mark

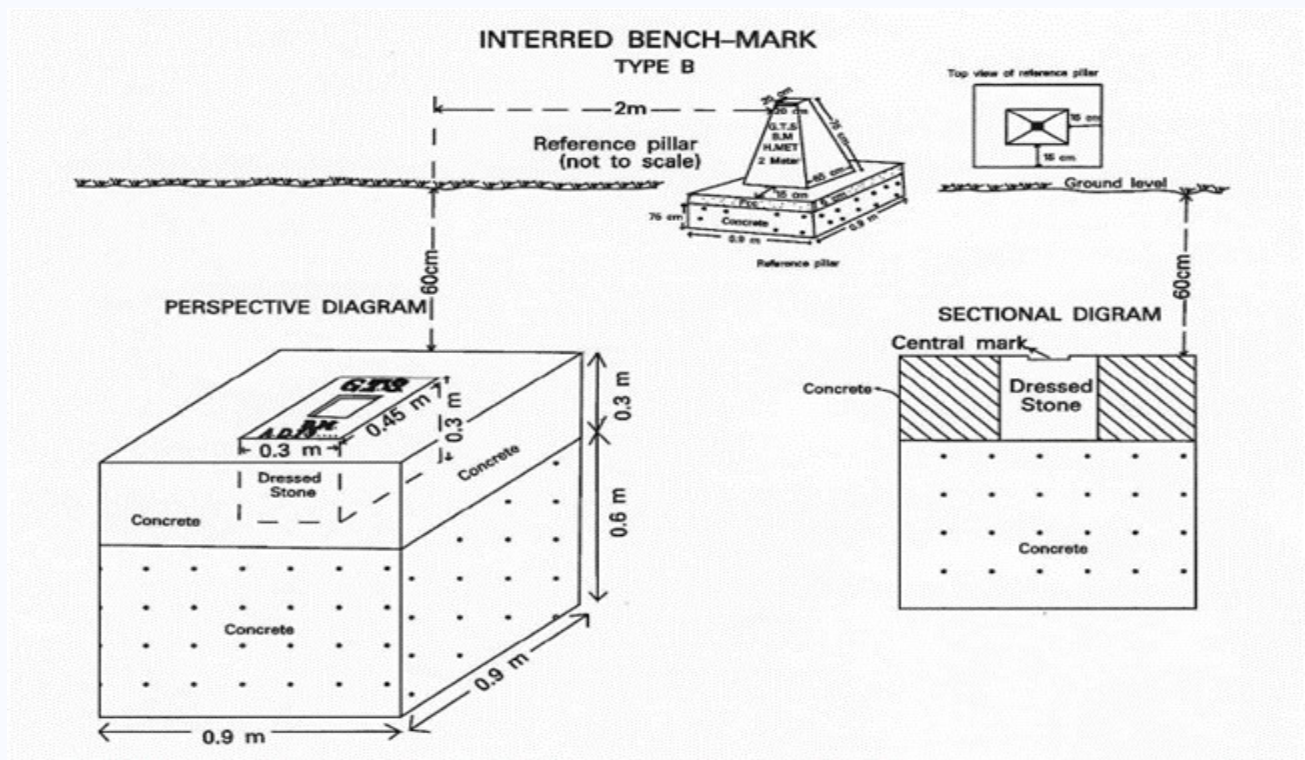


Figure 12: Diagram of Type 'B' Standard Bench Mark

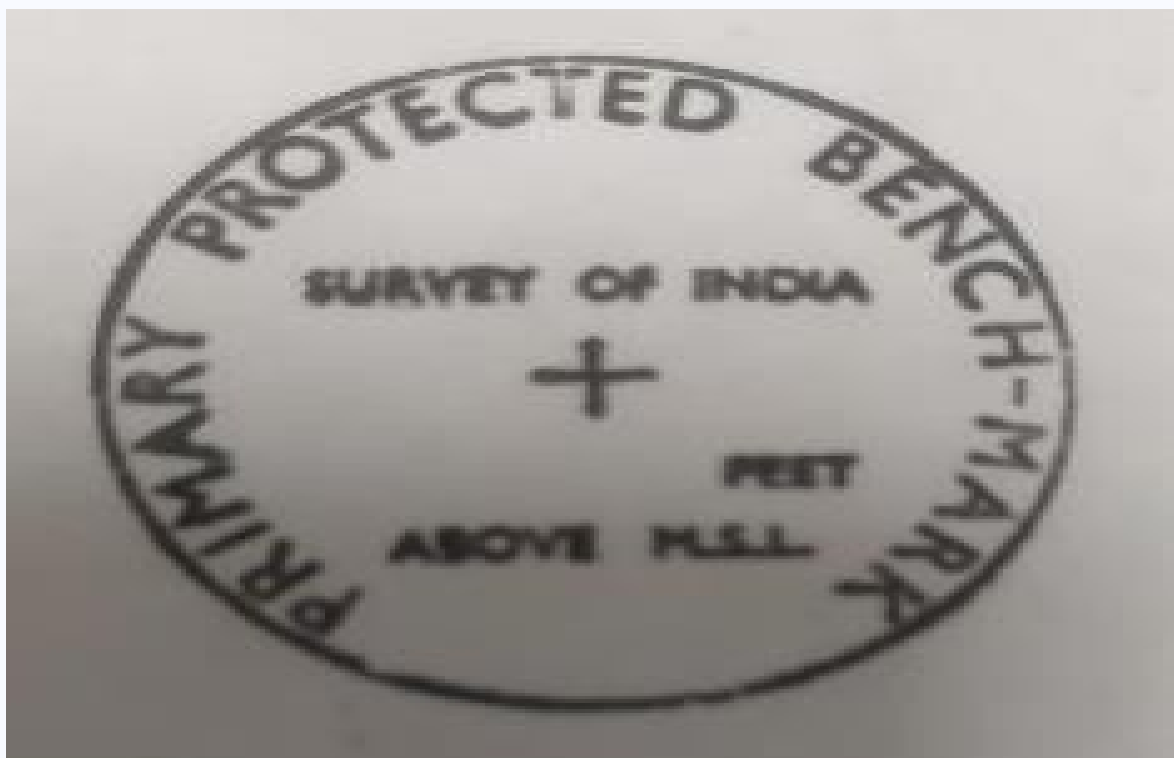


Figure 13: Type 'B' Standard Bench Mark

6 Standard Bench Mark Type 'C' (Rock-cut)

Important Note

Rock-cut bench marks are highly valuable as they can be created during field levelling operations.

Construction Details

- **Inscription:** "G.T.S. O B M" deeply engraved on solid ground rock
- **Dimensions:** Circle diameter about 5cm, letters of similar height
- **Protection:** Hollow masonry pillar 0.60m square, 0.30m high
- **Top stone:** Bears inscription "G.T. Survey O Upper Mark"



Figure 14: Type 'C' Standard Bench Mark



Figure 15: Photograph of Type 'C' Standard Bench Mark

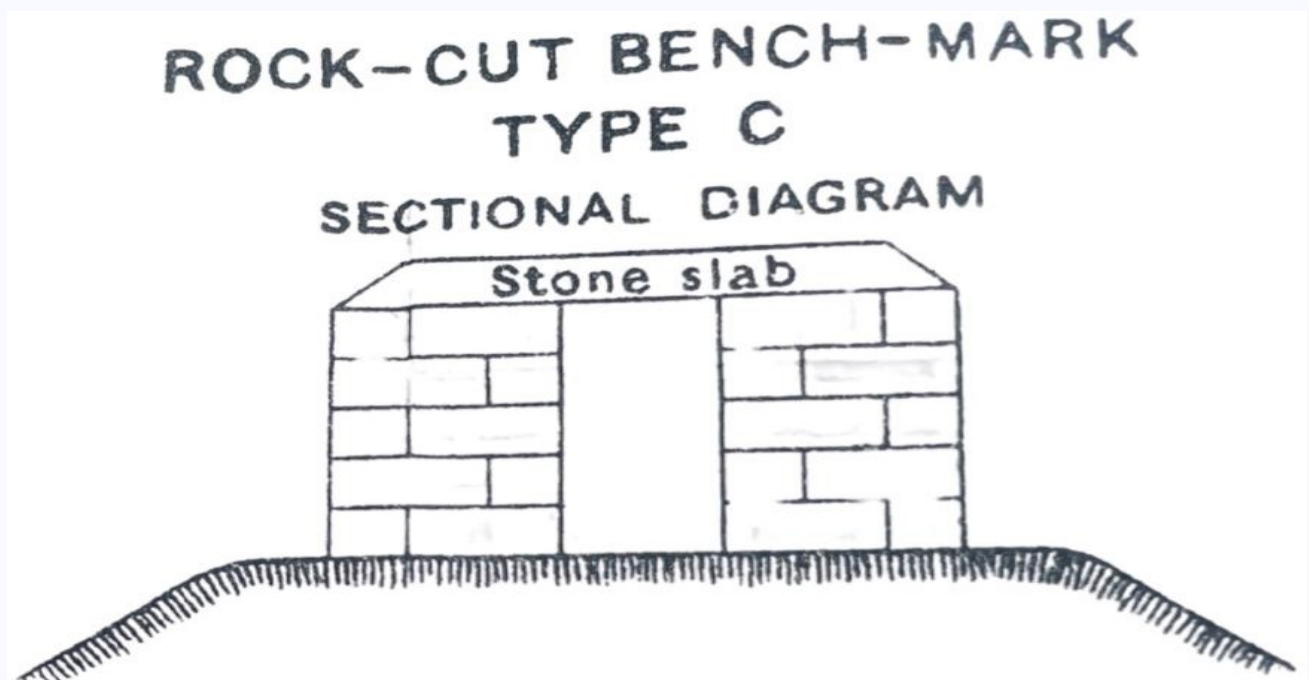


Figure 16: Diagram of Type 'C' Standard Bench Mark

7 Standard Bench Mark Type 'D'

Construction Details

- **Pillar:** Cement concrete, 0.45m square, 1.20m height (0.90m below ground)
- **Pipe:** Zinc pipe 2.5cm diameter, 1.20m length embedded flush with pillar surface
- **Alternative:** Hollow galvanized iron pipe if zinc unavailable



Figure 17: Photograph of Type 'D' Standard Bench Mark

8 Inscribed Ordinary Bench Marks (OBMs)

Important Note

Ordinary Bench Marks typically consist of the inscription pattern showing "G T S O B M" in various configurations, where "O" represents the measurement point. These are inscribed between two Standard Bench Marks with varying designs.

8.1 Common Locations for OBMs

- Ground rock
- Big boulder
- Memorials and tombs
- Church, shrine or temple
- Boundary pillar
- Ancient building
- Plinth or floor of government offices (post office, kacheri)
- Government circuit house/guest house
- Tanks and wells
- Railway stations and platforms
- Parapets of girder bridges
- Pedestals of lamp posts
- Plinth of culverts
- Plinth of KM stone pillars

References

1. Geodetic & Research Branch, Survey of India, Dehradun
2. Technical Instructions for Bench Mark Establishment